

OFFICE NETWORK BUILD

Ethernet & Switching

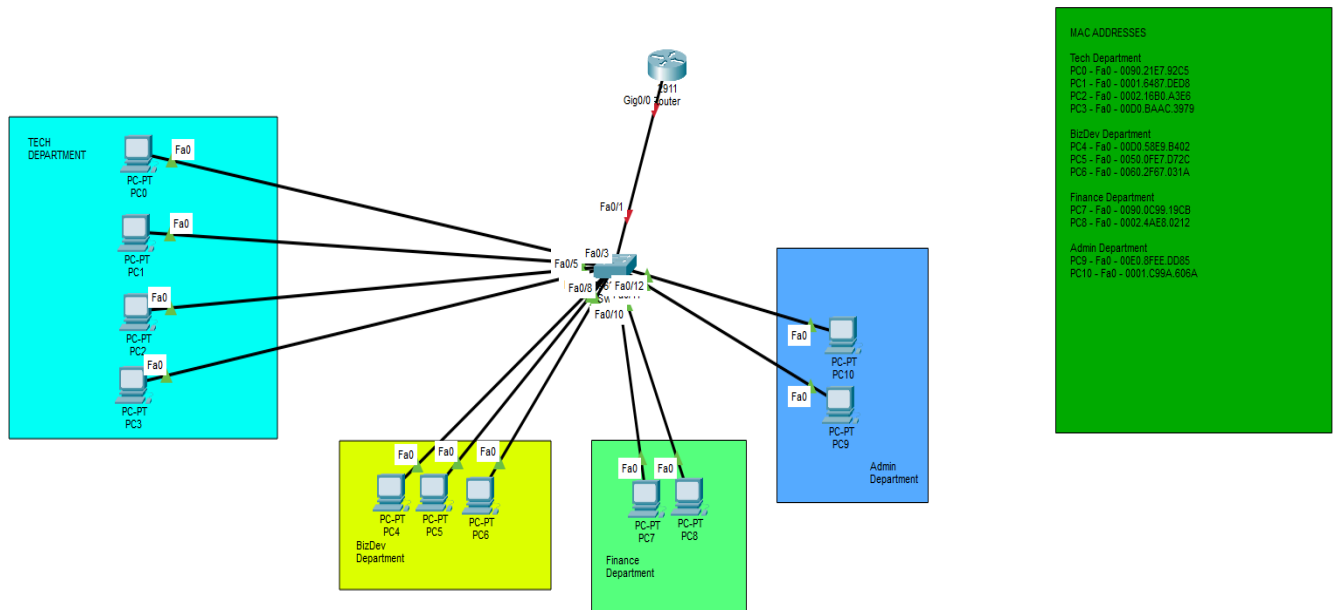
Following the completion of the IP subnetting phase of the Office Network Build, the next stage focuses on understanding how devices communicate within a local network through Ethernet and Layer 2 switching.

The network consists of four departments—Technology, Business Development, Finance, and Administration—with each department assigned its own IP subnet. Although the devices are logically separated through IP addressing, all endpoints are currently connected to a single Cisco switch. This provides an opportunity to examine how communication occurs at the Ethernet and Data Link layers before introducing technologies such as VLANs and inter-VLAN routing.

This project investigates the fundamental operations of an Ethernet switch, including MAC addressing, Ethernet frames, ARP, MAC address learning, frame forwarding, unknown-unicast flooding, and broadcast traffic. Using Cisco Packet Tracer, network traffic and switch behavior are observed and analyzed through practical experiments.

The objective is not only to verify that the network operates correctly, but also to understand how the switch makes forwarding decisions and how Layer 2 communication differs from Layer 3 routing. The findings from this stage will establish the foundation for the next phase of the Office Network Build, where logical network segmentation using VLANs will be introduced.

Topology:



```

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up

Switch>
Switch>en
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
Switch#

```

Initial MAC Address Table of Switch before MAC address learning. The table is empty because the switch has not yet learned any dynamic MAC addresses through Ethernet traffic. Subsequent communication between the PCs will allow Switch to learn and associate source MAC addresses with their respective switch ports.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128
Reply from 192.168.100.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.100.3

Pinging 192.168.100.3 with 32 bytes of data:
Reply from 192.168.100.3: bytes=32 time<1ms TTL=128
Reply from 192.168.100.3: bytes=32 time=7ms TTL=128
Reply from 192.168.100.3: bytes=32 time<1ms TTL=128
Reply from 192.168.100.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 1ms

C:\>ping 192.168.100.4

Pinging 192.168.100.4 with 32 bytes of data:
Reply from 192.168.100.4: bytes=32 time<1ms TTL=128
Reply from 192.168.100.4: bytes=32 time<1ms TTL=128
Reply from 192.168.100.4: bytes=32 time<1ms TTL=128
Reply from 192.168.100.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.100.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

```

%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/8, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/9, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/10, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/11, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/12, changed state to up

Switch>
Switch>en
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type      Ports
----    -
      1      0001.6487.ded8    DYNAMIC   Fa0/3
      1      0002.16b0.a3e6    DYNAMIC   Fa0/4
      1      0090.21e7.92c5    DYNAMIC   Fa0/2
      1      00d0.baac.3979    DYNAMIC   Fa0/5
Switch#

```

MAC Address Table After PC0 Connectivity Testing. After PC0 successfully pinged the other Tech Department PCs, Switch dynamically learned their MAC addresses and associated them with their respective switch ports. This demonstrates how a switch builds its MAC address table by examining the source MAC addresses of incoming Ethernet frames.

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.002	SW1	PC2	ARP
	0.002	SW1	PC5	ARP
	0.002	SW1	PC6	ARP
	0.002	SW1	PC7	ARP
	0.002	SW1	PC8	ARP
	0.002	SW1	PC9	ARP
	0.002	SW1	PC10	ARP
	0.003	PC6	SW1	ARP
	0.004	SW1	PC4	ARP
	0.004	--	PC4	ICMP
	0.005	PC4	SW1	ICMP
	0.006	SW1	PC6	ICMP
	0.007	PC6	SW1	ICMP
	0.008	SW1	PC4	ICMP
	1.008	--	PC4	ICMP
	1.009	PC4	SW1	ICMP
	1.010	SW1	PC6	ICMP
	1.011	PC6	SW1	ICMP
	1.012	SW1	PC4	ICMP
	1.455	--	SW1	STP
	1.456	SW1	PC0	STP
	1.456	SW1	PC8	STP
	1.456	SW1	PC9	STP
	1.456	SW1	PC10	STP
	1.456	SW1	PC2	STP
	1.456	SW1	PC3	STP
	1.456	SW1	PC4	STP
	1.456	SW1	PC5	STP
	1.456	SW1	PC6	STP
	1.456	SW1	PC7	STP
	1.456	SW1	PC1	STP

Reset Simulation ☒ Constant Delay

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, Profinet, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Captured to: 41.801 s

The Packet Tracer Simulation Panel shows the network activity generated during the connectivity test. Before the ICMP traffic is exchanged, ARP requests are observed being forwarded by Switch to multiple connected PCs, demonstrating the broadcast nature of ARP at Layer 2. Once the required MAC address information has been resolved, ICMP traffic is exchanged between the communicating hosts through Switch.

This observation demonstrates the relationship between ARP, Ethernet switching, and IP communication. ARP is used to resolve an IPv4 address to a MAC address, allowing the subsequent ICMP packets to be encapsulated in Ethernet frames and forwarded by the switch.

Conclusion

The Ethernet and Switching phase demonstrated how devices within the office network communicate at the Data Link and Network layers using Ethernet, MAC addressing, ARP, and Layer 2 switching. Through the experiments performed in Cisco Packet Tracer, the switch was observed dynamically learning MAC addresses from incoming frames and using its MAC address table to make forwarding decisions.

The simulation also demonstrated the role of ARP in resolving IPv4 addresses to MAC addresses before communication can occur. ARP requests were observed being broadcast across the local network, while subsequent ICMP traffic was forwarded toward the appropriate destination after the required MAC information had been learned.

These experiments provided a practical understanding of the difference between broadcast traffic, unknown-unicast flooding, and known-unicast forwarding, as well as how a switch operates within a Layer 2 network. The tests also reinforced an important distinction between IP subnetting and Layer 2 switching: although the four departments are physically connected to the same switch, their different IP subnets require Layer 3 routing for communication between them.

This phase establishes the Layer 2 foundation required for the next stage of the Office Network Build. With the behavior of Ethernet switching understood, the project can now progress toward VLANs, where the network will be logically segmented to provide greater control over the office's broadcast domains and departmental connectivity.